

Project Title:- ARES (Autonomous Rover for Explosive Safety)

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Abstract

Landmines continue to endanger military personnel and civilians in conflict zones. Manual detection and deployment are high-risk tasks that often result in casualties. To address this challenge, the project proposes the development of an autonomous ground rover (ARES) designed for landmine detection, mapping, and deployment. The rover integrates navigation systems, detection mechanisms, and a secure deployment unit, allowing it to operate in hazardous areas while minimizing human involvement. By combining automation, robotics, and communication technologies, ARES offers a safe, reliable, and efficient solution for modern defense operations.

Problem Statement

Landmine detection and deployment remain among the most dangerous military operations. Current methods expose soldiers to life-threatening risks and are slow, labor-intensive, and prone to human error. Additionally, existing detection systems are often limited in scope, lacking the autonomy to handle large minefields efficiently. There is an urgent need for a robust, autonomous, all-terrain system that can carry out these tasks effectively and safely.

Proposed Solution

- ARES is proposed as a semi-autonomous/fully autonomous rover capable of:
- Navigating through hazardous terrains using GPS and obstacle-avoidance systems.
- Detecting and mapping landmines with precision.
- Deploying landmines in predefined safe zones when required.
- Relaying real-time data and visuals to operators located at a safe distance.
- This system will significantly reduce soldier involvement in minefields, thus enhancing safety and operational efficiency for the Army.

System Workflow

The workflow of the ARES rover is as follows:

- Deployment – Rover is deployed into the field.
- Navigation – Uses GPS + obstacle sensors to move across terrain.
- Detection – Scans the ground and identifies landmines.

- Mapping – Marks GPS coordinates of mines in real time.
- Deployment – Places mines at designated spots.
- Monitoring – Sends live video and data back to the control station.
- Completion – Returns to base or shuts down safely.

Challenges

- Terrain Adaptability: Rover must operate in sand, mud, and rocky ground.
- Detection Accuracy: Ensuring reliable detection of both metallic and non-metallic mines.
- Power Management: Maintaining long operational life in the field.
- Communication Reliability: Secure and uninterrupted data transfer over long distances.
- Cost & Scalability: Balancing advanced features with affordability for large-scale deployment.

Expected Outcomes

- Working rover prototype capable of detecting and deploying landmines.
- Autonomous path planning and obstacle avoidance system.
- Real-time data transmission and remote operation interface.
- Demonstrated improvement in safety and efficiency compared to manual methods.

Conclusion

ARES represents a critical innovation in defence robotics, offering both strategic advantage and humanitarian value. By automating dangerous operations like mine detection and deployment, this project not only enhances operational efficiency but also saves lives. Its scalable and modular design makes it adaptable to various field conditions and missions.